

王业涵

电话: 152 8925 7841 | 邮箱: 10245102458@stu.ecnu.edu.cn

教育背景

华东师范大学

计算机科学与技术 | 本科 | GPA: 3.62/4.00

主修课程: 量子计算、数学分析、线性代数、离散数学、数据结构、算法分析与设计、数字逻辑电路、大学物理、概率论与数理统计、操作系统、机器学习等。

交流项目: 牛津大学暑期班 (师从Prof. Kobi Kremnizer, 修读《Quantum Computing》, 成绩97分) 2025.08

青年使者赴美(加州)交流项目 2025.01-2025.02

项目经历

医疗类AI产品 AdenoGuard——儿童腺样体智能筛查与辅助决策系统 | 第一开发人 2026.01-2026.03

- 真实临床数据与模型调优:** 在杨静教授、胡文心教授级工程师的指导和实验室成员的协同下, 依托三甲医院授权的10GB+真实多模态临床数据(既往病史、腔镜影像、鼾声音频), 开发多模态AI引擎以协助筛查腺样体肿大病症。核心搭载AutoGluon训练框架, AUC精度高达0.91。
- 端云协同架构与小程序开发:** 攻克跨模态数据流转难题, 实现多媒体医疗数据的无缝采集与全链路加密传输; 对接后端FastAPI接口, 保障云端AI推理快速响应; 使用SQL管理用户数据库。
- 医疗级闭环交互设计:** 将诊断结果通过CLAHE图像增强技术及梅尔谱图可视化呈现, 完成从结构化病史录入到治疗辅助决策的业务闭环。利用Canvas动态渲染含置信度仪表盘、风险折线趋势图的医疗报告单, 赋能无创居家早筛。
- 目前已产出体验版小程序, 下一步计划正式上线。

多模态智能地图生成与可视化排版系统 | 科研助理(导师: 李晨辉教授) 2026.02-至今

- 前后端数据对接与地图呈现:** 参与基于Next.js与FastAPI的系统搭建及Multi Agent管线部署, 学习并接入Mapbox GL引擎。
- 纯前端物理排版引擎落地:** 协助重构地图防遮挡布局算法, 在浏览器端引入KDE与力导向物理引擎。利用弹簧牵引与排斥力场动态演算, 实现卡片避让与防重叠交互。
- 系统联调与后续优化规划:** 在全链路联调中, 排查出大模型的跨省定位幻觉, 以及模型尝试手绘复杂SVG导致渲染卡顿和失真的问题。针对上述缺陷, 提出了引入地理围栏拦截异常坐标, 以及建立本地Icon库的优化思路。

竞赛经历

2026年 MCM/ICM 美国大学生数学建模竞赛 | 队长兼写作手 2026.01-2026.02

- 特征工程与数据重构:** 针对原始数据中分数归零、淘汰轮次缺失等问题设计标准化清洗策略; 通过分数归一化消除赛季尺度差异, 并基于领域知识创新性构造“The Kingmaker Effect”等高解释性特征。
- 模型构建与归因分析:** 综合运用XGBoost、随机森林与SLSQP优化算法构建混合模型; 利用回归任务量化明星背景特征的边际贡献, 结合SHAP进行特征归因, 解构并预测观众投票的隐藏偏好。
- 逆向优化与稳健性验证:** 构建逆向优化模型重构未知投票数据; 针对高不确定性核心变量设计敏感性分析, 揭示不同赛制在放大观众偏好上的结构性差异。
- 项目开源代码见github.com/genalyu/mcm-icm。

OS华东区域赛 | 核心队员 2025.12-2026.01

- 核心算法与机制设计:** 针对传统多级反馈队列易受CPU密集型进程“恶意欺骗调度”的漏洞, 设计并实现了基于智能行为预测的Smart MLFQ调度器。并引入指数加权移动平均算法, 使系统能动态识别并自适应多类型进程。
- 底层内核深度开发:** 深入uCore OS内核, 通过扩展PCB字段和schedule()调度, 实时采集I/O频率与CPU耗时等运行时特征, 有效保证各种类型的进程都有较好的调度结果。
- 系统压测与多维验证:** 构造纯CPU、纯I/O及混合型进程的压力测试以模拟真实运行环境。通过提取内核调试日志, 对比测算三种调度器的响应时间、周转时间与上下文切换次数, 全方位验证算法对系统吞吐量与公平性的提升。
- 重要代码详见GitLab: gitlab.eduxiji.net/T202510269998251/project3035746-358881。

荣誉奖项

- 2026年 美国大学生数学建模竞赛(MCM/ICM) 二等奖(Honorable Mention)
- 2026年 全国大学生计算机系统能力大赛-操作系统设计赛-华东区域赛-操作系统原理赛道 三等奖
- 2026年 第十五届“挑战杯”中国大学生创业计划竞赛华东师范大学校赛银奖
- 2025年 华东师范大学优秀本科生二等奖学金
- 2025年 华东师范大学优秀学生
- 2024年“外研社·国才杯”“理解当代中国”全国大学生外语能力大赛铜奖

专业技能

- 计算机能力:** C++、Python、SQL、JavaScript、HTML、CSS、微信小程序开发
- 语言能力:** 英语水平良好(高考英语143分, CET-6: 568分)
- 办公软件:** 熟练使用Word、PowerPoint、Excel、飞书

Yehan Wang

Mobile: 152 8925 7841 | E-mail: 10245102458@stu.ecnu.edu.cn

Education

East China Normal University

Shanghai, China

Bachelor of Computer Science | GPA: 3.62/4.00

Sep 2024 – Present

Relevant Courses: Quantum Computing, Mathematical Analysis, Linear Algebra, Discrete Mathematics, Data Structures, Algorithm Design and Analysis, Digital Logic Circuits, Physics, Probability and Statistics, Operating Systems, Machine Learning.

Exchange Programs:

·Oxford University Summer Institute – Quantum Computing (Advised by Prof. Kobi Kremizer) | Grade: 97/100 Aug 2025

·Youth Envoy Exchange Program – California, USA

Jan 2025 – Feb 2025

Projects

AdenoGuard – An AI-powered WeChat Mini Program for Pediatric Adenoid Hypertrophy Screening |

Lead Developer

Jan 2026 – Mar 2026

- **Real-world Clinical Data & Model Tuning:** Collaborated with Prof. Jing Yang and Prof. Wenxin Hu. Leveraged 10GB+ of real-world multimodal clinical data (medical history, endoscopic imaging, snoring audio) from a tertiary hospital. Built a multimodal engine using AutoGluon, achieving 0.91 AUC.
- **End-to-End Architecture & Mini Program Development:** Solved cross-modal data flow challenges. Implemented seamless multimedia data collection and encrypted transmission. Integrated with a FastAPI backend for low-latency cloud AI inference. Manage the user database using SQL.
 - **Medical-grade Closed-loop Interaction:** Visualized diagnostic results using CLARE image enhancement and mel-spectrograms. Built a complete business loop from structured medical history intake to treatment decision support. Dynamically generated medical reports (Canvas) featuring confidence dashboards and risk trend charts.

Multimodal Intelligent Map Generation & Visual Layout System | **Research Assistant (Advised by Prof.**

Chenhui Li)

Feb 2026 – Present

- **Data Integration & Map Rendering:** Participated in system development using Next.js and FastAPI, and deployed a Multi-Agent pipeline. Integrated the Mapbox GL engine.
- **Frontend Physics-based Layout Engine:** Contributed to refactoring the map anti-occlusion layout algorithm by introducing KDE and physics-based force-directed engines on the browser side.
- **System Debugging & Optimization:** Identified "cross-province positioning hallucinations" in LLMs and rendering freezes caused by complex SVG generation. Proposed using geofencing to filter anomalous coordinates and building a local icon library as solutions.

Contests

MCM/ICM 2026 | Team Leader & Writer

Jan 2026 – Feb 2026

- **Feature Engineering & Data Reconstruction:** Designed a standardized data cleaning strategy for issues like score resets and missing elimination rounds. Constructed high-interpretability features like "The Kingmaker Effect".
- **Model Building & Attribution Analysis:** Built an ensemble model using XGBoost, Random Forest, and SLSP optimization. Quantified marginal contributions and performed feature attribution analysis using SHAP to predict audience preferences.
- **Reverse Optimization & Robustness Validation:** Built a reverse optimization model to reconstruct unknown voting data and conducted sensitivity analysis.
- For more details on the project, please visit github.com/genalyu/mcm-icm.

CSCC 2025 Operating System Design Competition | Core Team Member

Dec 2025 – Jan 2026

- **Core Algorithm & Mechanism Design:** Designed and implemented a SmartMLFQ scheduler to mitigate "malicious deception" by CPU-intensive processes. Integrated the EWMA algorithm for dynamic process identification and adaptation.
- **Deep Kernel Development:** Extended PCB and modified the schedule function in the uCore OS kernel for real-time feature collection (I/O frequency, CPU time, resource utilization).
- **System Stress Testing & Validation:** Developed mixed workloads (CPU-bound, I/O-bound). Analyzed kernel debug logs to compare response time, turnaround time, and context-switch overhead, validating fairness and throughput improvements.
- For more details on the project, please visit gitlab.eduxiji.net/T202510269998251/project3035746-358881.

Honors & Awards

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| • MCM/ICM Honorable Mention | 2026 |
| • Third Prize – CSCC 2025 Operating System Design Competition (East China Region) | 2026 |
| • The 15th "Challenge Cup" China College Students' Entrepreneurship Plan Competition Silver Award | 2026 |
| • Second-class Excellence Scholarship – East China Normal University | 2025 |
| • Outstanding Student Award – East China Normal University | 2025 |
| • Third Prize – FLTRP "Understanding Contemporary China" National Foreign Language Proficiency Competition | 2024 |

Technical Skills

- Programming: C++, Python, Pytorch, JavaScript, WeChat Mini Program Development
- Languages: Proficient in English (Gaokao: 143/150, CET-6: 568)
- Office Software: Proficient in Word, PowerPoint, Excel, Lark